The error budget

Where does the error come from, and which terms are ours to control?

`Notebook companion: Unit-08-error-budget.ipynb`

1 The question

The previous seven units each produced numbers. This one asks where the error in those numbers comes from, term by term, and sorts the terms by who can do something about them.

Three layers contribute, and each is controlled by a different kind of work. The measurement itself is limited by photons and by the detector, and that layer is bought with hardware. The step from an optical depth to a column density is limited by how well the cross sections are known, and that layer is settled in the literature rather than on the bench. The step from columns to a profile is limited by whether the atmosphere is doing what the retrieval assumes, and that layer is an algorithm choice. Confusing the three is how a project ends up trying to solve a literature problem with a better detector.

The unit also carries a correction that we found while building it. Every precision figure this project has quoted was computed with a noise model that contains only one of the two random terms. Section 8b puts the second term in and reports what changes.

2 Physical picture

The instrument measures a ratio. I_0 is the solar flux through the channel with the ray above the atmosphere, I is the flux with the ray through it, and the optical depth is $\tau = -\ln(I/I_0)$. A fractional error ϵ in that ratio becomes an *absolute* error ϵ in τ, so the fractional error in the column is ϵ/τ. The same one per cent of radiometry costs five per cent of column at $\tau = 0.2$ and a quarter of a per cent at $\tau = 4$.

That single fact organizes the measurement layer. Four terms enter it. Two are random from one measurement to the next, so they set the scatter; two are biases with a sign, so they move every measurement the same way and do not average down.

Shot noise is the photon count. The signal falls as $e^{-\tau}$, so the relative photon noise grows as $e^{\tau/2}$. `occ.SHOT` is its value at $\tau = 0$ for one 10 km altitude bin.

Flat-field drift is the one the project has been leaving out. The pixel's own gain does not matter, because I_0 is measured through the same pixel at the same wavelength and the gain divides out. What does not divide out is the *change* in that gain between the reference frame and the occultation, which is why the vendor question asks for stability over a year rather than for uniformity. A one per cent drift is an error of 0.01 in τ, and unlike shot noise it does not care how deep the absorption is.

Out-of-band leakage is light that reaches the detector without passing through the filter's in-

tended band. It adds a floor to the measured signal, so the measured transmission is $T + f$ rather than T, and τ comes out too small. The error is negligible in the middle of the window and grows sharply where the atmosphere is nearly opaque, because there the leak is most of what is left.

Residual optical depth on the reference is the fact that I_0 is taken through the top of the atmosphere and not through vacuum. Unit 5 measured it: the aluminium channel still reads an optical depth of 0.019 from a 456 km host with the Sun on the local horizontal, falling to 0.006 with the Sun ten degrees up. Every τ is then too small by that amount.

Above the measurement layer sit two more terms. The cross section converts τ into a column, so an error in σ is an error in N; Unit 4 found that the published values carry ten to twenty per cent for stable species and that no N_2-specific figure exists at all for the predissociation region. And the tangent altitude is not measured but inferred, so an error in it shifts the whole profile; the column error that follows is $\delta z / H_n$, with H_n the density scale height of that species.

Finally there is the model-form term, which is not an error in any measured quantity at all. The four-parameter fit assumes diffusive equilibrium on a Bates temperature profile. When the atmosphere departs from that shape the fit returns a wrong answer with a small scatter, and section 8e measures how wrong.

3 Derivation

Start from the column error. Since $\tau = \sigma N$ in the thin limit and the band average of Unit 6 preserves the relation to first order,

$$\frac{\delta N}{N} \;=\; \frac{\delta \tau}{\tau} \;\oplus\; \frac{\delta \sigma}{\sigma},$$

and the four measurement terms enter the first part as

$$\delta\tau_{\mathrm{shot}} = s_0\, e^{\tau/2}, \qquad \delta\tau_{\mathrm{flat}} = \epsilon_f, \qquad \delta\tau_{\mathrm{leak}} = -\ln(1 + f e^{\tau}), \qquad \delta\tau_{I_0} = -\tau_0.$$

The first two add in quadrature and the last two add with their signs. Their sum has a minimum inside the window rather than at either wall, which is a design statement: the ladder should place channels near the deep end of the window, not the middle.

For registration, a shift δz changes the column of species s by

$$\frac{\delta N_s}{N_s} \;=\; \frac{\delta z}{H_{n,s}}, \qquad H_{n,s} \;=\; -\left(\frac{\mathrm{d}\ln n_s}{\mathrm{d}z}\right)^{-1},$$

where H_n is the *density* scale height, which differs from species to species. The consequence is that a pointing error is not absorbed into an overall density scaling: it changes the composition ratio too, by $\delta z\,(1/H_{\mathrm{N_2}} - 1/H_{\mathrm{O}})$.

The cross-section term needs one more distinction, because σ can be wrong in two ways that do not behave alike. Scale $\sigma_s(\lambda)$ by a constant at every wavelength and the forward model is unchanged if N_s is divided by the same constant. The error maps one-to-one into that species' column, exactly, and it touches no other species. No channel design improves on that: the accuracy with which the *level* of a cross section is known is a hard floor under the accuracy of the column. An error with structure *inside* the band cannot be absorbed that way, so the fit trades the species against each other instead, and section 8c shows what that costs.

4 Worked numbers

8a. The measurement terms across the window

	absolute error in τ				column error		
τ	shot	flat	leak	I_0	random	bias	total
0.20	0.0044	0.0100	−0.0012	−0.0060	5.47 %	−3.61 %	6.55 %
0.50	0.0051	0.0100	−0.0016	−0.0060	2.25 %	−1.53 %	2.72 %
1.00	0.0066	0.0100	−0.0027	−0.0060	1.20 %	−0.87 %	1.48 %
2.00	0.0109	0.0100	−0.0074	−0.0060	0.74 %	−0.67 %	1.00 %
3.00	0.0179	0.0100	−0.0199	−0.0060	0.68 %	−0.86 %	1.10 %
4.00	0.0296	0.0100	−0.0532	−0.0060	0.78 %	−1.48 %	1.67 %

Flat-field drift 1 %, out-of-band rejection 10^{-3}, I_0 residual $\tau_0 = 0.006$ (Sun ten degrees up). "Total" adds the random and bias columns in quadrature, which is what a single measurement would show; the bias does not average down over many of them.

Three readings.

The window is not flat, and its best point is not in the middle. The random error falls from 5.5 per cent at $\tau = 0.2$ to a minimum of 0.68 per cent at $\tau = 2.8$ and rises again to 0.78 per cent at $\tau = 4$. The bias has its own minimum at $\tau = 1.9$. Both put the sweet spot of the ladder between $\tau = 2$ and $\tau = 3$, deeper than we have been placing it.

Shot noise, the only term the simulator has modelled, is the smaller of the two random terms over most of the window. It overtakes the flat-field floor only at $\tau = 1.83$. At the shallow wall it is less than half of it.

The I_0 residual alone justifies the elevation requirement adopted on 6 September. At $\tau_0 = 0.019$, the value Unit 5 measured with the Sun on the local horizontal, the bias at $\tau = 0.2$ is 9.5 per cent — essentially the whole accuracy benchmark spent on one term, in a part of the window we intend to use. Taking the reference at ten degrees elevation brings it to 3.0 per cent, and taking it deeper into the window brings it below one.

8b. What the missing flat-field term costs

The flat-field figure has been in the design since Unit 6 — it is what set the bottom of the optical-depth window — but it has never been in `occ.noise_sigma`. Putting it in changes nothing else: same atmosphere, same seed, same tangent-altitude grid, same reporting altitudes.

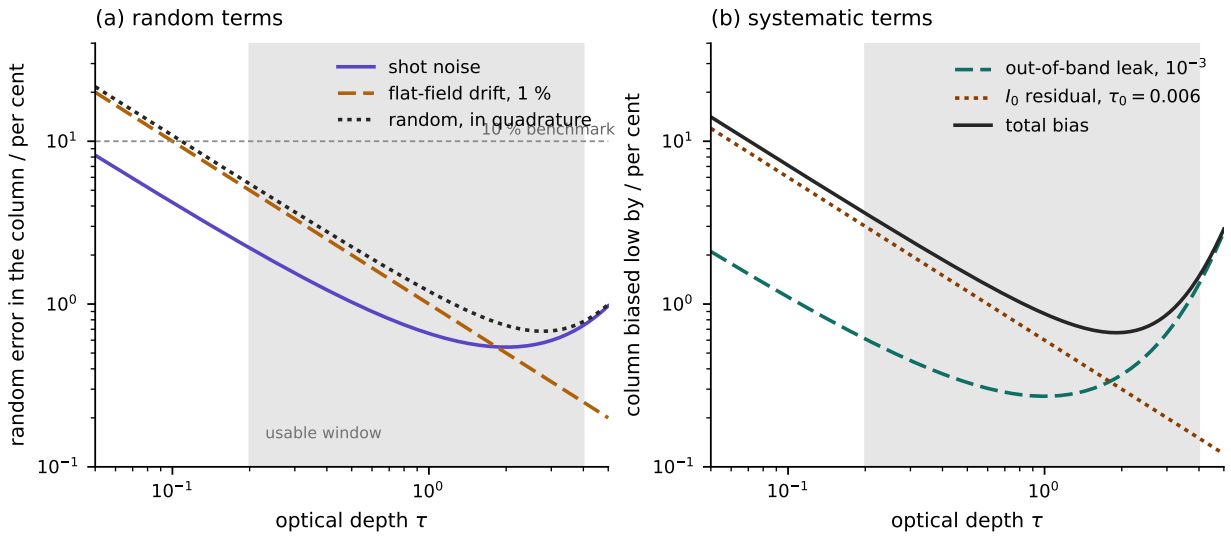


Figure 1: Column error from each measurement term against optical depth, with the usable window shaded. (a) The random terms. Shot noise dominates only at the deep end; the flat-field floor dominates everywhere else and is what makes the shallow wall expensive. (b) The systematic terms, both of which bias the column low. Generated by section A of the notebook.

Channels	pts	shot noise only	shot + flat field	ratio
quiet, lat 0, F10.7 140, ap 4				
Zr + Al	31	$+0.65 \pm 2.88$	$+0.93 \pm 4.38$	1.52
Zr + Al + In	46	$+0.72 \pm 0.80$	$+0.62 \pm 1.13$	1.41
Zr + Al + 2 FUV	37	$+13.41 \pm 2.15$	$+14.62 \pm 3.27$	1.52
all five	52	$+4.77 \pm 0.67$	$+4.05 \pm 0.98$	1.46
severe storm, lat 60, F10.7 200, ap 400				
Zr + Al	43	$+4.50 \pm 6.46$	$+7.83 \pm 12.42$	1.92
Zr + Al + In	67	-0.15 ± 1.45	-0.74 ± 2.43	1.68
Zr + Al + 2 FUV	62	$+11.40 \pm 5.49$	$+13.89 \pm 10.58$	1.93
all five	86	$+1.09 \pm 1.12$	$+1.50 \pm 2.02$	1.81

Bias and spread of retrieved atomic oxygen at 200 km, in per cent, O_2 held, 30 draws, 10 km tangent-altitude grid, Abel penalty 1.25 applied. "pts" is the number of channel–altitude points inside the τ window.

The second random term raises the spread by between 41 and 93 per cent, which is what the quadrature sum of section 8a predicts: the flat-field term is the larger of the two over most of the window, and the increase is biggest where the fit leans hardest on shallow points. The storm rows move most.

For the two-channel EUV configuration in a severe storm that is the difference between passing and failing. `STATUS.md` records 6.88 per cent for this case on 6 September; the larger ensemble here gives 6.46; with the second random term it is 12.42, and the benchmark is 10. This supersedes that entry, which was correct for the noise model it used and was quoted without saying what that model contained.

For the configurations that include indium it is not. Quiet, Zr+Al+In gives $+0.62 \pm 1.13$ per cent; in a severe storm, -0.74 ± 2.43. Both are comfortably inside the benchmark with the fuller noise model, and the reason is not the noise model at all but the number of independent measurements: indium adds a third channel that sees both species, and the fit has 67 usable points instead of 43.

8c. Cross-section knowledge

Two perturbations, both of +15 per cent in $\sigma(N_2)$, applied to the retrieval's copy of the cross section and not to the one that produced the observation.

	z	**uniform**		**6–20 nm only**	
Channels	km	O	N_2	O	N_2
Zr + Al	250	0.0	−13.0	−100	+14.0
	300	0.0	−13.0	−100	+27.1
	350	0.0	−13.0	−100	+39.2
Zr + Al + In	250	0.0	−13.0	−42.9	+1.1
	300	0.0	−13.0	−20.4	+2.1
	350	0.0	−13.0	−12.1	+2.4

Percentage error in the retrieved slant column, severe storm, noiseless optical depths, O_2 held at truth. −100 means the solve drove the oxygen column to zero.

The uniform case behaves exactly as the derivation says. A cross section fifteen per cent too large returns a column $1/1.15 - 1 = -13.04$ per cent too small, at every altitude, in every atmosphere, with two channels or three, and atomic oxygen is untouched to the last digit the solver carries. This is the floor: our N_2 columns cannot be more accurate than $\sigma(N_2)$ is known, and Unit 4 could find no published figure better than the ten to twenty per cent that Heays et al. (2017) give as the general range.

The structured case does not behave like that at all. Confine the same fifteen per cent to the zirconium band and, with two channels, the fit cannot absorb it into either column and instead drives oxygen to zero while inflating nitrogen. Adding indium reduces the same error to −12 to −43 per cent in oxygen and to two per cent in nitrogen. That is a third argument for the indium channel, independent of Unit 6's conditioning argument and Unit 3's storm-coverage argument.

The residual does not rescue us. With two channels and two unknowns the system is exactly determined, so a structured error is reproduced perfectly by a wrong pair of columns and leaves no residual to notice. The entries above show residuals only because the solve has collapsed against the edge of the parameter space, and even then the largest is 2.6 times the measurement error at 300 km and 1.0 times it at 250. A three-channel solve leaves one degree of freedom over and the residual rises to 3.5 times the measurement error — detectable, but only just, and only because the failure is already total.

There is a cost to that fix which should be stated in the same breath. Indium moves the retrieval's sensitivity into the 70–100 nm region, and that is exactly where Unit 4 found no N_2-specific uncertainty published at all. A fifteen per cent error confined to 80–100 nm barely touches the two-channel solve and produces +34 to +73 per cent in oxygen once indium is included. Indium does not remove

the cross-section sensitivity; it relocates it from a region that has been measured to one that has not.

8d. Tangent-altitude registration

	O		**N_2**		**O_2**	
z	H_n km	%/km	H_n km	%/km	H_n km	%/km
quiet						
180	39.1	2.56	23.3	4.29	19.2	5.21
200	42.9	2.33	25.1	3.98	21.4	4.68
250	47.2	2.12	27.1	3.69	23.6	4.23
300	48.7	2.05	27.8	3.59	24.4	4.11
severe storm						
180	49.2	2.03	32.4	3.08	28.8	3.48
200	59.1	1.69	38.0	2.63	33.8	2.96
250	79.5	1.26	48.9	2.05	43.3	2.31
300	93.9	1.06	56.0	1.79	49.4	2.02

Density scale height of each species and the column error produced by one kilometre of tangent-altitude registration error.

The binding species is O_2 in quiet conditions, at 5.2 per cent of column for every kilometre of registration error at 180 km. To hold the N_2 term below two per cent at 200 km the tangent altitude must be known to 0.50 km, which at a 3000 km slant range is 0.17 mrad. The detector row subtends 4.7 mrad (Unit 7), so this is registration to about one twentieth of a row.

That is not a number attitude knowledge will supply, and it should not be asked to. Occultation sounders register against the measurement itself — the shape of the transmission profile, or the edges of the solar disc crossing the slit — and the requirement above is what makes that a Phase 1 work package rather than a detail. It also cannot be absorbed into a density scaling: the species have different scale heights, so a registration error changes the composition ratio as well, by 1.65 per cent per kilometre in quiet conditions at 200 km.

8e. Composition, or how to be confident and wrong

`truth.composition_depletion` depletes oxygen and enhances nitrogen — the storm-time O/N_2 signature of Unit 3 — and changes nothing else, not even the temperature. The perturbation is a Gaussian centred at 200 km with a $1/e$ width of 50 km, so the factors in the table are its values at 200 km; by 300 km the shape has fallen to 0.018 of that, which is why the 300 km column of the retrieval barely moves and is not reproduced here. The retrieval is the four-parameter fit, which has no way to represent structure of this kind at all.

	O down	N_2 up	O at 200 km		O at 250 km	
d	%	%	bias	spread	bias	spread
1.00	0.0	0.0	+0.66	1.33	+0.23	1.46
1.05	4.8	1.7	+6.45	1.40	+1.91	1.49
1.10	9.1	3.3	+12.34	1.46	+3.54	1.51
1.25	20.0	8.3	+30.62	1.66	+8.08	1.57
1.50	33.3	16.7	+63.01	2.03	+14.78	1.66
1.75	42.9	25.0	+97.80	2.44	+20.77	1.75
2.00	50.0	33.3	+136.79	2.72	+25.65	1.88
2.50	60.0	50.0	+220.60	3.73	+36.27	2.09

Percentages. "O down" is $1 - 1/d$ and "N_2 up" is $(d - 1)/3$, both at the peak of the Gaussian. Zr+Al+In, O_2 held, 30 draws, 10 km grid, full noise model, storm base (lat 60, F10.7 200, ap 200).

The bias at 200 km crosses the ten per cent benchmark when oxygen is reduced by 7.4 per cent. That is a small disturbance by the standards of Unit 3, which found the O/N_2 ratio at 400 km falling by a factor of two from quiet to $ap = 400$. At the crossing point the retrieval reports a spread of 1.44 per cent, so the bias is seven times the uncertainty the retrieval quotes for itself, and at $d = 1.25$ — oxygen down twenty per cent — the ratio is 18.

Nothing in the fit signals the problem. The residuals stay small, the ensemble stays tight, and the retrieval returns a confident answer that is wrong by three times the benchmark. This is the single most important number in the unit, and it is why slant columns and not profiles are the primary product.

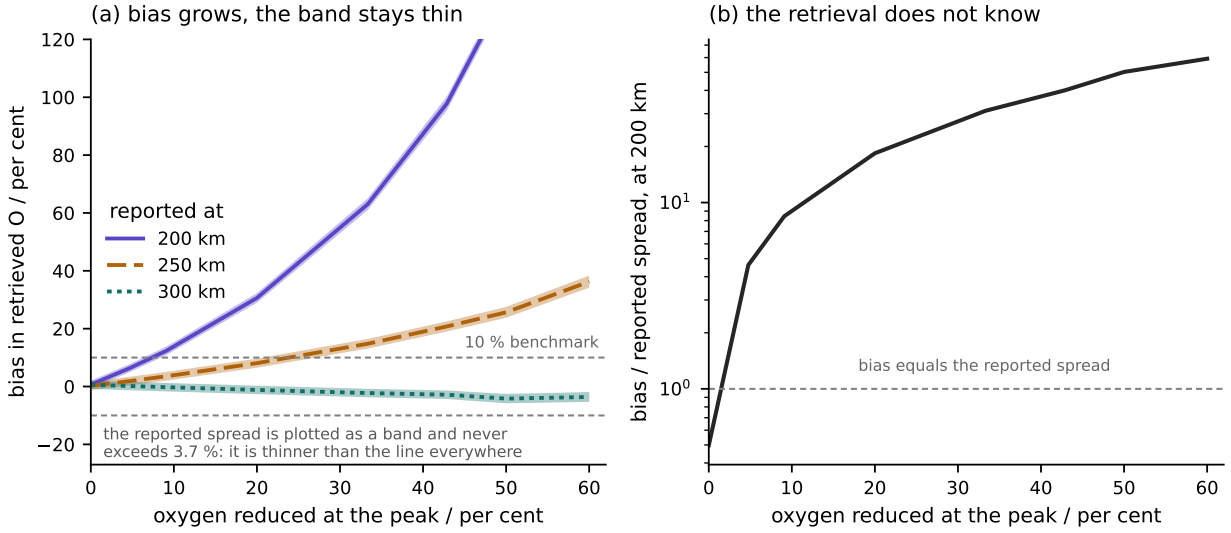


Figure 2: (a) Bias in retrieved atomic oxygen against the strength of the composition depletion, with the shaded band showing the spread the retrieval reports. The band stays thin while the curve leaves the benchmark. (b) The ratio of the two: where the bias reaches the benchmark it is already seven times the quoted uncertainty, and it climbs past sixty for the strongest depletion in the sweep. Generated by section E of the notebook.

8f. Temperature from hydrostatic integration

Unit 7 described `onion.hydrostatic_temperature` and did not measure its error. This is that measurement: the onion peel run on the five-channel set with a correct prior, temperature integrated downward from a top boundary estimated from the local scale height.

z	below top	T true	T retrieved	bias	spread
km	km	K	K	%	K
quiet					
395	5	838	827	−1.3	0
375	25	838	23330	+2683	0
355	45	838	1057	+26.1	187
315	85	838	689	−17.8	49
275	125	835	745	−10.8	26
255	145	833	785	−5.8	17
215	185	818	804	−1.7	17
175	225	764	762	−0.3	21

Twelve draws, prior equal to truth, retrieval top at 400 km, full noise model. The storm case is in the notebook: the same test gives an unbroken run from 175 to 255 km, but with a spread of 334 K at the top of it, falling to 98 K at 195 km.

Two things are wrong at the top and they are different problems.

The 23,000 K entry is not a temperature. It is what happens when the peel returns a shell density that is essentially zero: at 375 km in this run the retrieved oxygen came back at 3×10^{-7} of truth and at 365 km at 1×10^{-7}, the total density at three and four per cent of truth, and $T = p/(nk)$ divided by it. The temperature step did not fail; it displayed a failure of the density retrieval above it, which is Unit 7's noise gain of 110 for the topmost shells seen from another direction.

Below that the error is the top-boundary pressure decaying downward as advertised. Counting up from the bottom — the honest test, since an isolated good shell above a bad one is not a range anyone can report — the quiet bias stays under ten per cent in an unbroken run from 175 to 265 km, which is everywhere from 135 km below the top of the retrieval downward, and it reaches −0.3 per cent 225 km below. The oxygen scale height there is about 48 km, so the rule of thumb is closer to three scale heights than to two.

WHY NOT O_2 is held, and when it need not be

Holding a species means fixing it at its prior, and Unit 7 showed that a held species transfers its prior error into the species that are solved. So why hold O_2 at all? Because the baseline two-channel EUV configuration cannot see it. Freeing O_2 there takes the oxygen spread in a severe storm from ±12.4 to ±39.8 per cent, and O_2 itself comes back at ±140 per cent: an unconstrained parameter does not sit still, it wanders and drags its neighbours with it.

That is a statement about the channel set, not about O_2. Add indium and freeing it costs almost nothing — the oxygen spread goes from ±2.4 to ±3.6 per cent and O_2 is determined to ±2.0. With all five channels it is free of charge: ±2.02 against ±2.10, and O_2 itself to ±0.7 per cent. The rule we adopt is therefore conditional rather than absolute. O_2 is held only where no in-window channel absorbs it, and the retrieval publishes which species were solved at each altitude, exactly as Unit 7 required.

8g. Which configuration survives the whole budget

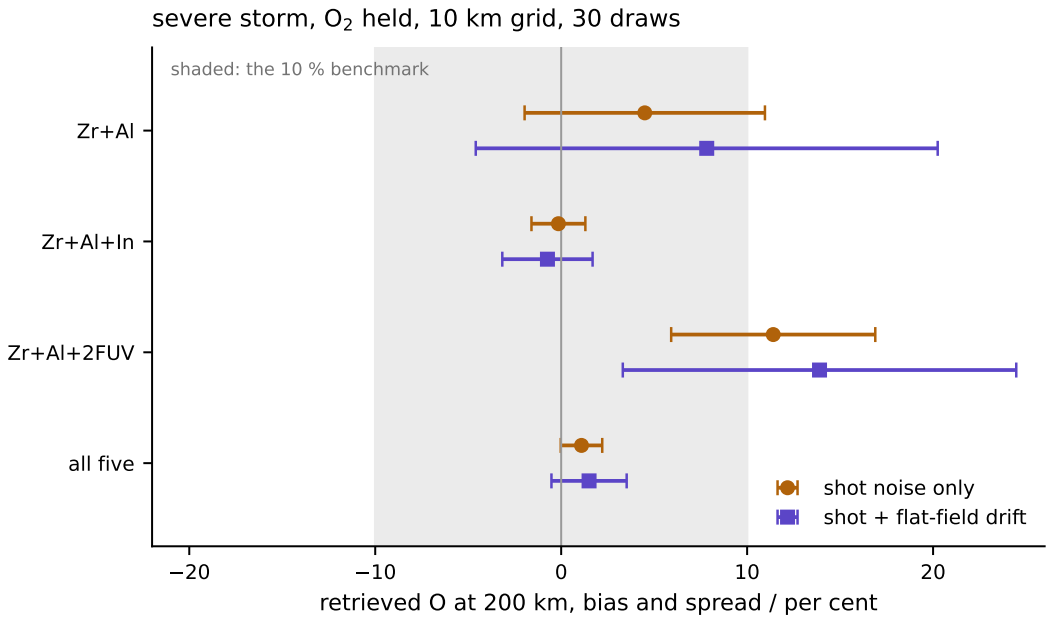


Figure 3: Retrieved atomic oxygen at 200 km in a severe storm under the two noise models, for the four channel sets, with O_2 held. The flat-field term widens every error bar by half again; the choice of channel set moves the centre. Generated by section G of the notebook.

The pattern in the table of section 8b is worth stating on its own, because it bears on a decision the project has already made.

Indium is what meets the benchmark. In a severe storm, under the full noise model, Zr+Al+In returns oxygen at -0.74 ± 2.43 per cent where Zr+Al returns $+7.83 \pm 12.42$. The 5 September decision records indium as an enhancement with Zr+Al+2 FUV as the fallback; on these numbers the fallback returns $+13.89 \pm 10.58$ and fails both halves of the benchmark, quiet as well as storm.

The same effect explains why the full five-channel set is no better than three in quiet conditions: $+4.05 \pm 0.98$ per cent against $+0.62 \pm 1.13$. Adding channels is not automatically adding information.

The reason the far-ultraviolet fallback is worse than the two channels it was meant to improve on

is not a hardware property, and it is worth understanding before anyone acts on the number. With O_2 held at its prior, the two FUV channels contribute residuals that depend only on the held O_2 and on the two shape parameters, so they act as a strong constraint on the temperature profile, and that constraint pulls the oxygen density with it. Freeing O_2 more than halves the bias, from +13.89 to +6.24 per cent. The FUV channels are not useless; they are useless *held*.

5 What the computation shows

The project's noise model was missing its larger random term. Shot noise dominates only above $\tau = 1.83$; below that the flat-field drift is larger, and at the shallow wall it is more than twice as large. Adding it raises the spread of every retrieval by 41 to 93 per cent, and it takes the two-channel EUV storm case from 6.5 to 12.4 per cent, across the benchmark.

A uniform cross-section error is a floor and a structured one is a hazard. Scaling σ everywhere maps exactly into the column of that species and touches nothing else, so ten to twenty per cent of literature uncertainty is ten to twenty per cent of column accuracy and no design improves it. An error confined to part of the band cannot be absorbed that way: with two channels it drives the oxygen column to zero while leaving a residual smaller than the noise.

Registration is a tighter requirement than it looks. One kilometre of tangent-altitude error is 2.3 per cent of the oxygen column and 5.2 per cent of the O_2 column, and because the species have different scale heights it distorts composition as well as density. Half a kilometre is one twentieth of a detector row.

The model-form term is the one that hides. Every other term in this unit announces itself, in the scatter or in the residual. A seven per cent reduction in oxygen puts the retrieved oxygen outside the benchmark while the retrieval reports a spread of 1.4 per cent.

And the channel set decides whether any of this matters. Indium turns the storm case from failing on both accuracy and precision into passing on both, makes a structured cross-section error survivable, and makes freeing O_2 affordable. It also moves the residual cross-section sensitivity into the one spectral region where nobody has published an uncertainty.

6 What we decided

`occ.noise_sigma` gains the flat-field drift; `occ.shot_sigma` preserves the old model so that any recorded number can be reproduced. The out-of-band leak and the I_0 residual are budgeted as biases in section 8a rather than added to the noise, because that is what they are. Every precision figure the project quotes is restated against the fuller model. The affected row of `STATUS.md` is the precision benchmark, which moves from 6.5 to 12.4 per cent for the two- channel EUV configuration in a severe storm and to 2.4 per cent with indium. The old figure was not wrong for the model it used; it was quoted without saying what that model contained, which is the thing to fix.

Every quoted error carries the noise model, the channel set, the tangent-altitude grid and the atmosphere that produced it. Section 8b is four numbers for the same instrument, and only the labels separate them.

Cross-section knowledge is quoted as two numbers from now on, a level and a structure, because they propagate differently. The literature task in `STATUS.md` is narrowed accordingly: what is

needed is not one uncertainty for $\sigma(N_2)$ but a statement of how the uncertainty varies across 6–100 nm, and in particular whether the 80–100 nm predissociation region is known to better than fifteen per cent.

Tangent-altitude registration to 0.5 km becomes a stated instrument requirement, met from the measurement rather than from attitude knowledge, and the method for doing so is a Phase 1 work package alongside the hybrid retrieval.

Hydrostatic temperature is reported only from about three scale heights below the top of the retrieval downward — 135 km in the quiet case measured here — and the retrieval publishes the altitude at which that cut falls.

One item goes back to KC rather than being decided here. The 5 September decision treats indium as an enhancement and Zr+Al+2 FUV as an equivalent fallback. Section 8g says the fallback is not equivalent under this budget. That does not reopen the channel set — indium is already in the baseline — but it does change what happens if Luxel cannot supply it, and it should be settled before the 3 October sensor decision rather than after.

7 What we do not claim

Not that the flat-field term is correctly modelled as independent noise. Within one occultation the treatment is right: successive tangent altitudes fall on different detector rows, and their gain errors are unrelated. Across occultations it is not, because the same rows recur. The spread quoted above is therefore the profile-to-profile spread one would see if the pattern were redrawn each time; a real instrument has one pattern, so what it will actually show is a repeatable offset of about that size, which averaging more occultations does not remove. The correct treatment carries the pattern as a nuisance parameter, and it is not in this budget.

Not that one per cent is measured. It is the number in RFI question 2. Every figure in section 8b scales with it, and a vendor answer of two per cent would double the flat-field term again.

Not that the leak model is right. A single grey rejection fraction is a placeholder. Real out-of-band leakage has spectral structure, and a leak that lands where the atmosphere is transparent behaves very differently from one that lands where it is opaque.

Not that the cross-section perturbations are the true uncertainties. Fifteen per cent over named intervals is a probe chosen to sit inside the ten-to-twenty per cent range Heays et al. report. It is not a measured covariance and the entries in section 8c should not be read as error bars.

Not that the temperature failure at the top is fundamental. It is the onion peel's density error in the topmost shells propagating into the pressure integration. A better cap estimate, a regularized peel, or simply not reporting the top three shells would each change it.

Not that the budget is complete. Missing, and each capable of mattering: pointing jitter within an integration, detector nonlinearity, scattered light from the bright limb below the tangent point, the top-hat filter approximation itself, and horizontal structure along the ray. The last of these is the largest known omission and Unit 5 has already measured the length of ray it applies to.

8 Exercises

E8.1 Section 8a puts the best point of the optical-depth window between $\tau = 2$ and $\tau = 3$, and the channel ladder was built to place channels in the middle of the window. Compute where each of the five channels actually sits in τ at 200, 250 and 300 km in quiet conditions, and say whether moving the ladder deeper would improve the budget or simply lose altitude coverage. State which of the two you would rather give up.

E8.2 The flat-field term enters the fit as independent noise across altitude, which section 7 of "What we do not claim" argues is right within an occultation and wrong across occultations. Modify the Monte Carlo so that the flat-field realization is drawn once and reused for every draw, run the storm case both ways, and report the difference between the ensemble spread and the offset of the ensemble mean. Say which of the two a user assimilating these profiles would care about.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.